

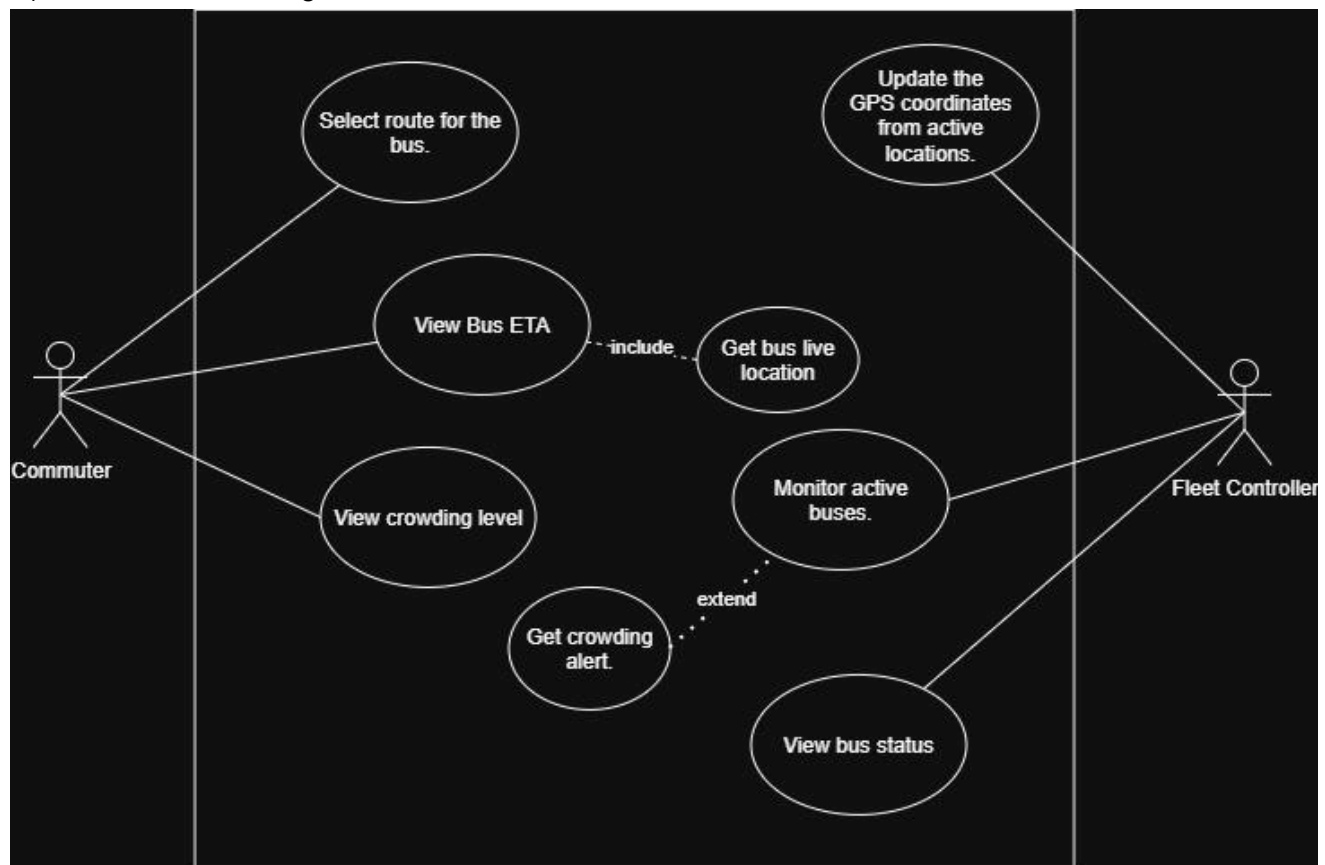
SE LAB 01
RYONA ADITREE GHOSH
PES1UG24AM233
AIML 5D

1) Requirements:

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall get GPS coordinates from active city buses every 5 seconds and update their current locations.	High	The GPS contains updated data of all the current locations.	Required for preparation .
FR-002	Functional	The system shall calculate and display estimated arrival times (ETA) for buses at upcoming route stops.	High	ETA is calculated based on current traffic information and road information and displayed to customers.	Commuters need accurate information about the ETA of buses.
FR-003	Functional	The system shall compute passenger crowding levels using data received from ticketing sensors.	High	The system calculates and displays the current crowding level for an active bus.	Crowd information helps commuters choose suitable buses and helps controllers monitor capacity.
FR-004	Functional	The system shall allow a commuter to	High	A commuter can select a bus or route and view its current location,	Provides commuters with the live

		view live bus locations, ETA information, and crowding levels for buses and route stops.		ETA, and crowding level.	transport information they need.
FR-005	Functional	The system shall allow the Fleet Controller to monitor active buses and their operational status through the tracking platform.	High	The Fleet Controller can view active buses and their latest tracking information.	Fleet monitoring supports effective management of the municipal bus service.
NFR-001	Non-Functional	The tracking engine shall handle telemetry streams from up to 1,000 active transit vehicles simultaneously.	High	Benchmarking confirms that the system continues processing telemetry within the required performance limits for 1,000 active vehicles.	The platform must remain usable during peak periods with many active buses.
NFR-002	Non-Functional	The system shall protect bus telemetry and passenger-related sensor data from unauthorized access or modification.	High	Access-control and security tests confirm that unauthorized users cannot access or modify protected data.	Protecting operational and passenger-related data is essential for a secure transit platform.

2) UML Use-Case Diagram:



3) Use-Case Flow Specification:

Main Success Scenario

1. The commuter opens the Public Bus Live Tracking & Crowding Estimator platform.
2. The commuter selects a route or bus.
3. The system displays the selected bus and its upcoming route stops.
4. The commuter selects the desired stop.
5. The system retrieves the latest GPS location of the selected bus.
6. The system calculates the estimated arrival time using the current bus location, distance to the stop, and available traffic information.
7. The system displays the calculated ETA to the commuter.
8. The commuter views the ETA and uses the information to plan their journey.

Alternate Flow

A1. GPS data is temporarily unavailable

1. At Step 5, the system detects that the selected bus has not provided recent GPS data.
2. The system cannot calculate a reliable current ETA.
3. The system informs the commuter that the ETA is unavailable.
4. When updated GPS data is received, the system recalculates the ETA.
5. The updated ETA is displayed to the commuter.

Preconditions

1. The commuter has access to the Public Bus Live Tracking & Crowding Estimator platform.
2. The selected bus is active and transmitting GPS data.
3. The bus route and upcoming stops are available in the system.

Postconditions

1. The commuter is shown the estimated arrival time of the selected bus at the chosen stop.
2. The displayed ETA reflects the latest available bus location and traffic information.

Actors

1. Commuter
2. Fleet Controller